

Lecture #11: The Likelihood Function & Maximum-Likelihood Estimators

MATH 321 – Mathematical Statistics

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The Likelihood Function

Suppose we collect data from a known distribution, but we don't know the parameters. How can we use this data to **estimate** the true value of the parameters?

For example, suppose I'm polling for tomorrow's gubernatorial election in Virginia between the Democrat Abigail Spanberger and the Republican Winsome Earle-Sears.

I can treat each voter's as a Bernoulli random variable, but how can I estimate the true proportion of voters who will vote for Spanberger?

The Likelihood Function

Let $X = [X_1, \dots, X_N]^T$ be a random vector drawn from a joint PDF $f_X(x; \theta)$, and let $x = [x_1, \dots, x_N]^T$ be the realizations. The **likelihood functions** is a function of the parameter θ given the realizations x :

$$\mathcal{L}(\theta | x) \stackrel{\text{def}}{=} f_X(x; \theta).$$

Given i.i.d. random variables $X_1, \dots, X_N$ that all have the same PDF $f_{X_n}(x_n)$, the **likelihood function** is:

$$\mathcal{L}(\theta | x) \stackrel{\text{def}}{=} \prod_{n=1}^N f_{X_n}(x_n; \theta).$$

What are the constants in these equations? What are the variables?

The Log-Likelihood Function

Given a set of i.i.d. random variables $X_1, \dots, X_N$ with PDF $f_{X_n}(x_n; \theta)$, the **log-likelihood** is defined as

$$\log \mathcal{L}(\theta \mid x) = \log f_X(x; \theta) = \sum_{n=1}^N \log f_{X_n}(x_n; \theta).$$

The **log-likelihood** is useful because it turns a product into a sum, which makes it much easier to differentiate, but the critical points remain the same before and after taking the logarithm.

Example: The Log-Likelihood

Example: Find the log-likelihood of a sequence of i.i.d. Bernoulli random variables $X_1, \dots, X_N$ with parameter θ .

Example: Find the log-likelihood of a sequence of i.i.d. Normal random variables $X_1, \dots, X_N$ with mean μ and variance σ^2 .

The Maximum-Likelihood Estimator

Let $\mathcal{L}(\theta)$ be the likelihood function of the parameter θ given the measurements $x = [x_1, \dots, x_N]^T$. The **maximum-likelihood estimate of the parameter θ** is a parameter that maximizes the likelihood:

$$\hat{\theta}_{ML} \stackrel{\text{def}}{=} \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta | x).$$

Example: Maximum-Likelihood Estimators

Example: Find the ML estimate for a set of i.i.d. Bernoulli random variables $\{X_1, \dots, X_N\}$ with $X_i \sim \text{Bernoulli}(\theta)$ for $i = 1, \dots, N$.

Example: Find the ML estimates for a set of i.i.d. Normal random variables $\{X_1, \dots, X_N\}$ with $X_i \sim \text{Normal}(\mu, \sigma^2)$ for $i = 1, \dots, N$.

Estimate vs. Estimator

An **estimate** is a **number**, e.g. $\hat{\theta}_{ML} = \frac{1}{N} \sum_{i=1}^N x_i$. It is the random realization of a random variable.

An **estimator** is a **random variable**, e.g. $\hat{\Theta}_{ML} = \frac{1}{N} \sum_{i=1}^N X_i$. It takes a set of random variables and generates another random variable.

Unbiased Estimators

An estimator $\widehat{\Theta}$ of θ is **unbiased** if

$$E[\widehat{\Theta}] = \theta.$$

Example: Let $X_1, \dots, X_N$ be i.i.d. Normal random variables with unknown mean μ and unknown variance σ^2 . We have shown that the ML estimators are:

$$\hat{\mu}_{ML} = \frac{1}{N} \sum_{i=1}^N X_i \quad \text{and} \quad \hat{\sigma}_{ML}^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \hat{\mu}_{ML})^2.$$

It is easy to show that $E[\hat{\mu}_{ML}] = \mu$. How about $\hat{\sigma}_{ML}^2$? Is it an unbiased estimator?

Consistent Estimators

An estimator $\widehat{\Theta}$ of θ is **consistent** if $\widehat{\Theta}_N \xrightarrow{p} \theta$, i.e.,

$$\lim_{N \rightarrow \infty} P \left[\left| \widehat{\Theta}_N - \theta \right| \geq \varepsilon \right] = 0.$$

Example: The sample mean $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$ is a consistent estimator of $E(X)$. Why?

Example: The sample variance is a consistent estimator of $V(X)$. Why?